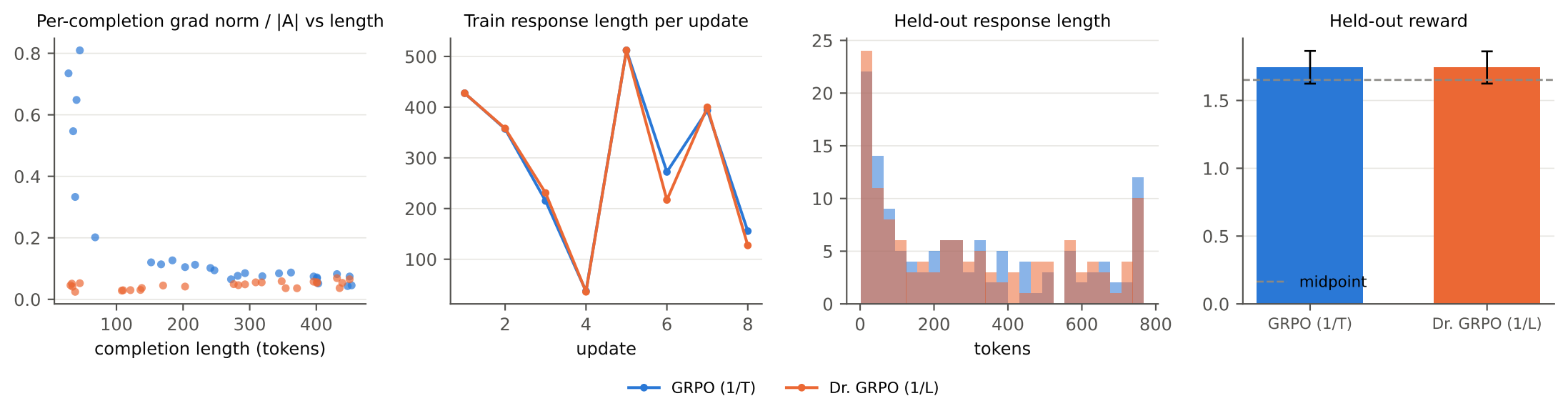
Per-completion grad norm / |A| vs length
0.0
0.2
0.4
0.6
0.8
100
200
300
400
completion length (tokens)
Train response length per update
100
200
300
400
500
2
4
6
8
update
Held-out response length
0
5
10
15
20
25
0
200
400
600
800
tokens
Held-out reward
0.0
0.5
1.0
1.5
midpoint
GRPO (1/T)
Dr. GRPO (1/L)
GRPO (1/T)
Dr. GRPO (1/L)